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SYSTEMATIC REVIEW

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Prevalence and associated factors of hypertension among adults in Sub-Saharan Africa: a systematic review and meta-analysis of community-based studies

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Abstract

Introduction Hypertension (HT) is a major global public health challenge, responsible for more than 10 million deaths annually. In sub-Saharan Africa (SSA), its prevalence is high and rising, driven by rapid urbanisation, nutritional transition, and increasing sedentary behaviour, while screening and control rates remain suboptimal. This systematic review and meta-analysis aimed to estimate the prevalence of HT among adults in SSA and identify its key associated factors.

Methods The protocol was registered in PROSPERO (CRD420251021429) and conducted in accordance with PRISMA 2020. We included cross-sectional studies published between January 2010 and November 2025, involving adults (≥ 18 years) and defining HT according to JNC7. Searches were conducted across seven databases and grey literature. Methodological quality was assessed using Joanna Briggs Institute tools (threshold ≥ 50%). A random-effects meta-analysis was performed using Stata 16.

Results Of 11,895 records screened, 52 studies were included. The pooled prevalence of HT was 27.09% (95% CI: 24.22%–29.96%; I^2 = 99.5%). Regional estimates ranged from 22.43% in Southern Africa to 32.75% in Central Africa. Major factors associated with HT included age > 50 years, which conferred nearly a fivefold increased risk (OR = 4.93; 95% CI: 3.61–6.72), male sex (OR = 1.80; 95% CI: 1.14–2.84), diabetes mellitus (OR = 4.01; 95% CI: 2.88–5.58), abdominal obesity (OR = 2.86; 95% CI: 2.33–3.52), overweight/obesity (OR = 2.72; 95% CI: 2.26–3.28), alcohol consumption (OR = 1.97; 95% CI: 1.57–2.46), smoking (OR = 1.69; 95% CI: 1.44–1.97), physical inactivity (OR = 1.60; 95% CI: 1.36–1.89), low fruit and vegetable intake (OR = 1.42; 95% CI: 1.16–1.75), urban residence (OR = 1.21; 95% CI: 1.08–1.35), and a family history of HT (OR = 2.37; 95% CI: 1.66–3.38). Occupational status was not significantly associated with HT (OR = 1.21; 95% CI: 0.76–1.93).

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Conclusion More than one in four adults in SSA is hypertensive, with substantial regional disparities. Most identified determinants are modifiable, underscoring the urgent need for strengthened primary prevention through healthier lifestyles, community-based screening, and integrated management of metabolic disorders. Context-specific strategies are essential to curb this silent epidemic and reduce the cardiovascular burden across the region.

Keywords Hypertension, Sub-Saharan Africa, Prevalence, Associated factors, Systematic review, Meta-analysis, Community-based studies, Cardiovascular risk, Epidemiology, Adults

Introduction

Hypertension (HT) remains one of the leading contributors to global cardiovascular morbidity and mortality, accounting for more than 10 million deaths each year, mainly through stroke, ischaemic heart disease, heart failure and kidney disease [1]. Its global burden has risen substantially over recent decades, from 594 million adults in 1975 to 1.28 billion in 2019, two-thirds of whom live in low- and middle-income countries (LMICs) [2, 3]; this figure reached 1.4 billion in 2024 [4].

Sub-Saharan Africa (SSA) exemplifies the epidemiological transition occurring across many LMIC settings. HT prevalence in the region ranges from 20% to 40% depending on the sub-region [1, 5, 6]. In the WHO African Region, the number of hypertensive adults aged 30–79 years increased from 45 million in 1990 to 106 million in 2019, corresponding to a regional prevalence of 36% [3]. The care cascade highlights substantial gaps: in 2019, only 43% of hypertensive individuals were diagnosed, 27% treated and 12% controlled—the lowest control rate across all WHO regions [3]. These deficiencies in screening, treatment and control contribute to a largely preventable cardiovascular burden [7].

This growing burden is driven by interconnected factors including population ageing, rapid urbanisation, sedentary lifestyles, high salt intake, and increased consumption of processed foods. Smoking, alcohol use, calorie-dense diets and accelerated urbanisation further heighten cardiovascular risk, underscoring the urgent need for prevention and management strategies adapted to SSA contexts [5, 7–12].

In response, the Pan-African Society of Cardiology (PASCAR) has proposed a 10-point action plan aiming to reduce HT prevalence by 25% by 2025, emphasising community screening, equitable access to essential medicines and strengthened primary care services [13, 14]. However, implementation remains limited due to structural barriers such as restricted access to healthcare, high treatment costs, low health literacy and persistent behavioural factors [15, 16].

Despite this growing burden, data on HT in SSA remain fragmented. Existing studies are often limited by small sample sizes, substantial regional heterogeneity and methodological inconsistencies, undermining comparability and generalisability [17, 18]. Moreover, prior reviews present several shortcomings that justify an updated meta-analysis. Some focus exclusively on older adults [12], others examine uncontrolled HT or populations with comorbidities [19], while several are restricted to specific subregions of SSA [20, 21]. In addition, many reviews rely on data published before 2015 [22], limiting their relevance to current epidemiological trends. To date, no synthesis has comprehensively assessed community-based HT among the general adult population across the period 2010–2025 or incorporated the most recent risk profile dynamics.

These gaps highlight the need for an up-to-date meta-analysis to yield robust estimates of HT prevalence and associated factors, which are essential for informing targeted prevention and control strategies in SSA settings. In this context, a systematic review and meta-analysis is warranted to provide an updated estimation of the prevalence and key determinants of HT among adults in SSA. This evidence aims to guide data-driven health policies and support implementation of the PASCAR action plan, thereby contributing to Sustainable Development Goal targets on reducing non-communicable diseases.

Materials and methods

Protocol and registration

This systematic review was conducted in accordance with the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [23]. The protocol was prospectively registered on PROSPERO under the registration number CRD420251021429.

Eligibility criteria and information sources

Studies were eligible if they included adult participants aged 18 years or older residing in SSA and applied a standardised definition of HT according to the Seventh Report of the Joint National Committee (JNC7), defined as systolic blood pressure ≥ 140 mmHg and/or diastolic blood pressure ≥ 90 mmHg, or current use of antihypertensive medication, regardless of measured blood pressure [24]. Eligible studies were required to report prevalence data and/or associated factors, employ a cross-sectional, cohort, or case-control design, be published between January 2010 and November 2025 without language restrictions, and be conducted in community settings encompassing urban, rural, or mixed populations. Although data on hypertension awareness, treatment, and control were extracted when available, reporting across studies was highly inconsistent, and definitions varied substantially. Due to the limited number of studies providing comparable information, meta-analysis of these indicators was not feasible, and they were therefore not included in the quantitative synthesis.

Studies were excluded if they focused exclusively on participants under 18 years of age, investigated secondary HT due to specific organ or endocrine disorders, were conducted in hospital or clinical settings, or targeted non-representative subgroups such as students, military personnel, or employees of specific industries. Publications were also excluded if they did not report quantitative data on HT prevalence or associated factors, or if they were non-primary research, including randomised controlled trials, systematic reviews, meta-analyses, qualitative studies, letters, editorials, or conference abstracts.

Discrepancies between reviewers were resolved through full-text assessment. For studies with missing data, up to two attempts were made to contact the corresponding authors by email before excluding the study.

Search strategy

A comprehensive literature search was conducted across multiple electronic bibliographic databases, including PubMed/MEDLINE, Web of Science, Scopus, Embase, Google Scholar, African Journals Online (AJOL), and Wiley Online Library. The search was further extended to grey literature sources such as OpenGrey, Google Scholar for theses, dissertations, and unpublished communications, as well as relevant institutional and government reports, to maximise completeness and minimise publication bias.

The systematic search strategy combined MeSH terms and free-text keywords related to hypertension, prevalence, risk factors, and Sub-Saharan Africa. English-language search terms included, among others, "hypertension," "blood pressure," "prevalence," "risk factors," "associated factors," "Sub-Saharan Africa," "obesity," "diabetes," "urbanisation," and "lifestyle." Search queries were adapted to each database, and the search was restricted to studies published between January 2010 and May 2025. Reference lists of included articles were manually screened to identify additional relevant studies.

Study selection

The review focused exclusively on SSA countries and included only quantitative, full-text studies, in accordance with the PRISMA 2020 guidelines [23]. References retrieved were managed using the Covidence platform, which facilitated tracking of studies at each stage, duplicate removal, and data organisation. The initial literature search was conducted between 1 January and 30 April 2025, and updated in November 2025 to incorporate the most recent publications. The full-text screening process was illustrated using a PRISMA 2020 flow diagram to ensure transparency and reproducibility.

Study selection was conducted independently by three reviewers using a rigorous two-stage process. Initially, titles and abstracts were screened for potential eligibility, followed by a detailed full-text assessment of preselected articles to confirm inclusion or exclusion based on predefined criteria. Discrepancies between reviewers were resolved by consensus, and persistent disagreements were adjudicated by a third reviewer.

For studies with missing critical data or essential methodological information, up to two attempts were made to contact the corresponding authors via email. If no response was obtained, the study was excluded from the analysis.

Quality assessment, risk of bias, and data extraction

The methodological quality of included studies was appraised using the Joanna Briggs Institute (JBI) critical appraisal tools, tailored to the specific study design (cross-sectional, cohort, or case-control) [25]. The assessment focused on eight key aspects: clear definition of participant inclusion criteria, detailed description of the study population and setting, validity and reliability of exposure measurement tools, clarity of objectives and selection criteria, identification of potential confounders, strategies employed to control for confounding, validity and reliability of outcome measurement, and appropriateness of the statistical analysis.

Each study received a binary score for each criterion (1 = yes, 0 = no or unclear), which was then converted into a percentage. Studies were categorised as low quality (≤ 49%), moderate quality (50–69%), or high quality (≥ 70%). Only studies scoring ≥ 50% were retained for analysis.

Data were extracted using a structured Microsoft Excel template, capturing the following information: author, year of publication, country and region, geographical setting (urban or rural), study objective, study design, population characteristics (sample size, mean age, sex distribution), HT definition, reported prevalence, and examined associated factors. In cases of missing or unclear data, two attempts were made to contact the corresponding authors via email to clarify or complete the information.

Statistical analysis

Extracted data were synthesised narratively and quantitatively, depending on the availability and comparability of data. Where pooling was feasible, meta-analyses were conducted using Stata version 16 (StataCorp, College Station, TX, USA) with a random-effects model, given the anticipated methodological and contextual heterogeneity between studies. Event counts, non-events, prevalences, and odds ratios (ORs) were initially recorded in Microsoft Excel and then imported into Stata. Weighted prevalences and pooled ORs with 95% confidence intervals (CIs) were calculated to estimate the meta-analysed burden and factors associated with HT.

Inter-study heterogeneity was assessed using Cochran's Q test and quantified with the I^2 statistic, interpreted as low (25%), moderate (50%), or high (75%) heterogeneity, with a p-value < 0.05 indicating statistically significant heterogeneity. Given that some analyses showed extremely high heterogeneity (I^2 > 99%), we emphasize that pooled prevalence estimates should be interpreted with caution, as a single summary measure may not fully capture the variability across studies. To address this, prediction intervals were calculated to reflect the range of expected prevalence in individual study settings, providing a more meaningful interpretation of the pooled results in the context of high heterogeneity.

For meta-analyses of associated factors, the Hartung-Knapp-Sidik-Jonkman method was applied to obtain more robust combined effect estimates and heterogeneity measures. Subgroup analyses were performed where data permitted, according to relevant criteria such as geographical sub-region, setting (urban vs. rural), and study period, to explore potential sources of variation in the pooled estimates. Publication bias was evaluated through visual inspection of funnel plots and Egger's regression test, with $p < 0.05$ indicating significant asymmetry and potential bias. To improve normality and stabilise variance, a logit transformation of prevalences was applied before publication bias tests. Finally, the main results were presented in forest plots, providing a concise visual summary of point estimates and 95% CIs for both meta-analysed prevalence and associated factors.

Results

Characteristics of included studies

The literature search yielded a total of 11,895 references, of which 11,841 were identified through electronic databases: PubMed (*n* = 8,958), Web of Science (*n* = 2,051), Embase (*n* = 397), African Journals Online (*n* = 209), Google Scholar (*n* = 159), and Scopus (*n* = 80). An additional 54 references were identified from other sources, including citation tracking (*n* = 35) and grey literature (*n* = 16). The PRISMA flow diagram (Fig. 1) details the study selection process.

Of the 11,841 database records, 627 duplicates were removed (7 manually and 619 via Covidence), leaving 11,268 records for title and abstract screening. One study was automatically excluded by automation tools. A total of 980 full-text articles were assessed for eligibility, and 52 studies met all inclusion criteria and were retained for the systematic review. All included studies (*n* = 52) were cross-sectional in design. Consequently, all prevalence estimates were calculated directly from the population-level data reported in each study, ensuring methodological consistency.

Among the 52 included studies, most were conducted in Ethiopia (13 studies) [26–38], followed by Nigeria (8 studies) [39–46], Kenya (4 studies) [47–50], and Ghana [52–54] and Uganda [51–56], each contributing three studies. Several countries were represented by two studies: Benin [57, 58], Cameroon [59, 60], the Democratic Republic of the Congo [61, 62], Malawi [63, 64], Sudan [65, 66], Tanzania [63, 67], and Zambia [68, 69]. One study originated from Angola [70], Burkina Faso [71], The Gambia [72], Madagascar [73], Mozambique [74], Rwanda [63], Senegal [75], South Africa [76], and the Central African Republic [77]. The sub-regional distribution was uneven, with the Eastern region contributing the largest number of studies (*n* = 24), followed by the Western region (*n* = 16). In contrast, the Central and Southern regions were substantially underrepresented, with only six studies each. The geographical distribution of the studies is presented in Fig. 2. All included studies were community- or subnational population-based surveys conducted in urban, rural, or mixed settings. Although two studies [68, 72] covered large geographic areas and were labelled as national.

This meta-analysis included a total of 169,857 participants. Sample sizes across the included studies ranged from 518 [32] to 28,891 [64]. Overall, 41,964 individuals were identified as hypertensive, with the number of hypertensive cases per study ranging from 108 [30] to 4,318 [72].

Quality assessment

The methodological quality of the 52 included studies was assessed using the Joanna Briggs Institute (JBI) critical appraisal checklist for analytical cross-sectional studies. Each criterion was rated as "yes" or "no," and an overall score was calculated as the percentage of positive responses.

No study scored below 50%. Three studies achieved a score of 63%, thirteen studies scored 75%, and the majority (*n* = 35) reached 88%, indicating generally good methodological quality. Two studies attained the maximum score of 100%. The criteria least frequently satisfied related to the identification and control of confounding factors, suggesting a potential risk of confounding bias in several studies. Nevertheless, the majority of studies provided adequate descriptions of participants and study settings, employed reliable and valid measurement tools, and conducted appropriate statistical analyses (Table 1).

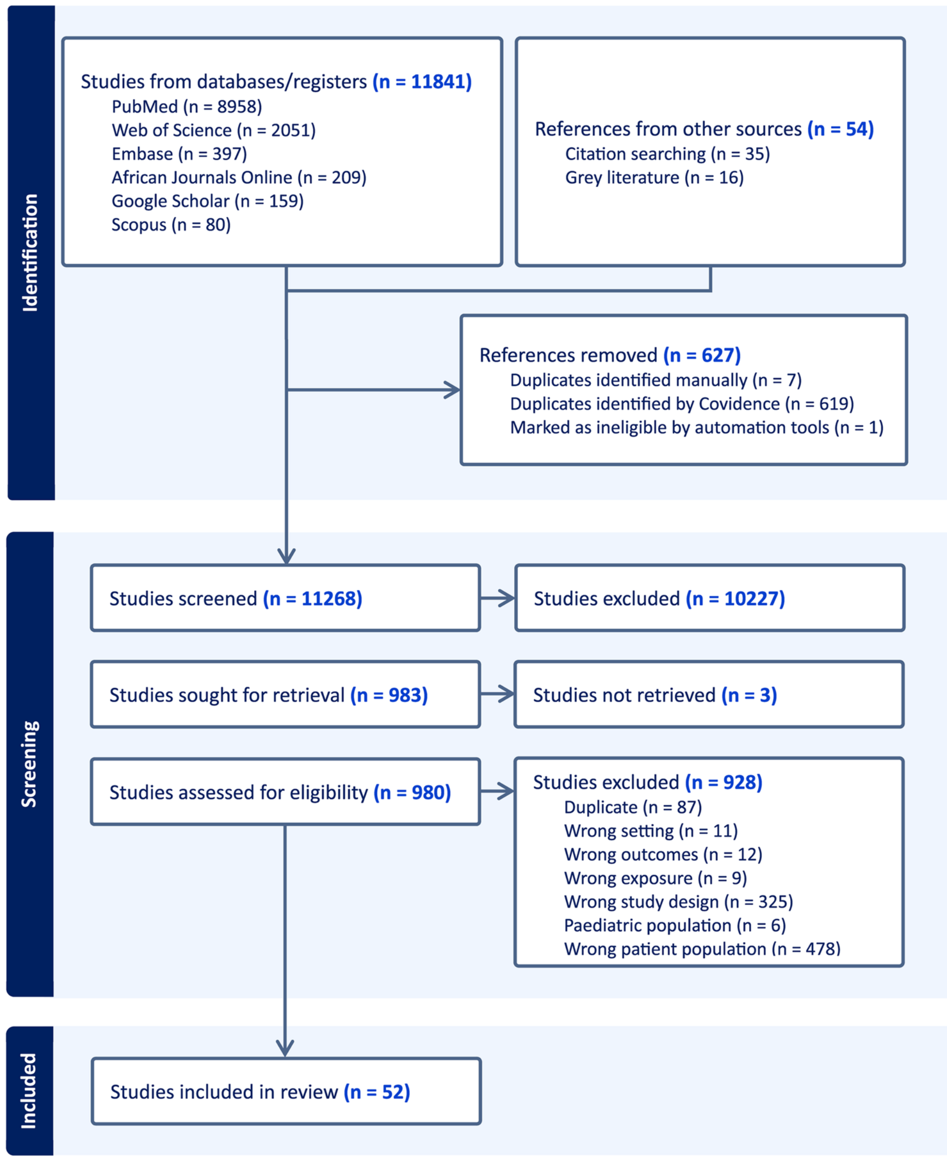

Fig. 1 PRISMA flow diagram illustrating the study selection process for the systematic review

Meta-analyses

Meta-analysed prevalence of hypertension among adults in Sub-Saharan Africa

Our meta-analysis of 52 studies estimated the meta-analysed prevalence of HT among adults in SSA at 27.09% (95% CI: 24.22%–29.96%). Heterogeneity was extremely high, with an I^2 of 99.53% and a τ^2 of 110.02, confirmed by a significant Cochran's Q test ($p < 0.0001$). Prevalence estimates varied widely between studies, ranging from 11.39% [38] to 57.65% [60], reflecting substantial epidemiological heterogeneity across the region (Fig. 3).

Regional analyses revealed particularly high prevalence in Central Africa (32.75%, 95% CI: 20.85%–44.65%) and West Africa (30.66%, 95% CI: 24.74%–36.58%), whereas prevalence was more moderate in East Africa (24.45%, 95% CI: 21.38%–27.53%) and Southern Africa (22.43%, 95% CI: 15.73%–29.14%). Heterogeneity remained extreme across all regions (I^2 > 99%), indicating substantial intraregional variation. The between-group test (Q = 5.59, $p = 0.134$) showed no statistically significant differences in prevalence across sub-regions of SSA (Figure S1).

Analysis by setting revealed notable variations (Figure S2). Meta-analysed prevalence was highest in mixed (urban/rural) settings at 29.98% (95% CI: 25.44%–34.52%), followed by rural areas at 25.11% (95% CI: 18.94%–31.27%) and urban areas at 23.18% (95% CI: 20.03%–26.34%). Heterogeneity remained very high within each group (I^2 > 98%), and differences between settings were not statistically significant (Q = 5.81; $p = 0.055$).

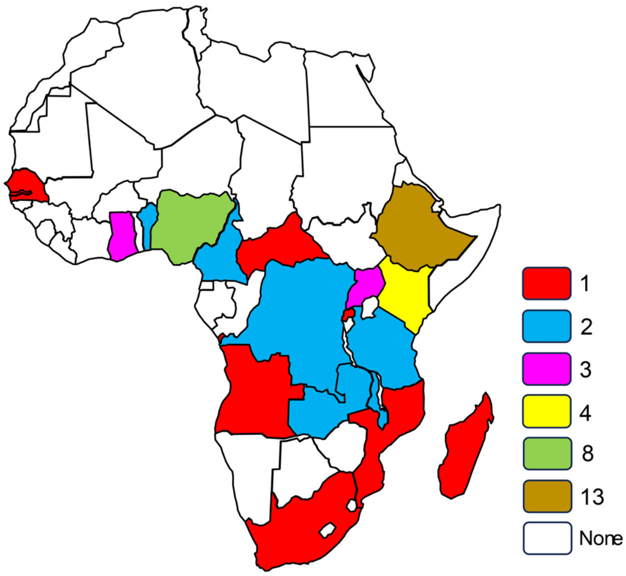

Fig. 2 Geographical distribution of studies included in the systematic review

Stratification by study period showed no statistically significant differences (Q = 0.72; p = 0.70). The meta-analysed prevalence was 24.26% (95% CI: 20.31%–28.12%) for studies conducted between 2010 and 2014, 29.17% (95% CI: 25.25%–33.09%) for 2015–2019, and 29.15% (95% CI: 17.50%–40.80%) for 2020–2024 (Figure S3).

An Egger's regression test was conducted to assess potential publication bias, which could indicate a small-study effect. Following logit transformation of the prevalence estimates, the results did not reveal statistically significant publication bias (β_1 = − 1.74; p = 0.5687), suggesting funnel plot symmetry and the absence of a small-study effect (Fig. 4).

Associated factors for hypertension among adults in Sub-Saharan Africa

This meta-analysis indicates that individuals aged over 50 years have nearly a fivefold higher risk of developing HT compared with those under 50 (OR = 4.93; 95% CI: 3.61–6.72). Urban residence is also associated with increased risk (OR = 1.21; 95% CI: 1.08–1.35). Regarding sex, men have a higher likelihood of HT than women (OR = 1.80; 95% CI: 1.14–2.84), whereas occupational status shows no significant association (OR = 1.21; 95% CI: 0.76–1.93) (Figure S4).

Alcohol consumption nearly doubles the risk of HT (OR = 1.97; 95% CI: 1.57–2.46), and low fruit and vegetable intake is similarly linked to increased risk (OR = 1.42; 95% CI: 1.16–1.75). Physical inactivity and smoking raise HT risk by 60% (OR = 1.60; 95% CI: 1.36–1.89) and 70% (OR = 1.69; 95% CI: 1.44–1.97), respectively (Figure S5).

A family history of HT more than doubles the risk (OR = 2.37; 95% CI: 1.66–3.38), and diabetes mellitus is strongly associated, with a fourfold increased risk (OR = 4.01; 95% CI: 2.88–5.58). Abdominal obesity and overweight/obesity increase HT risk by approximately 2.9-fold (OR = 2.86; 95% CI: 2.33–3.52) and 2.7-fold (OR = 2.72; 95% CI: 2.26–3.28), respectively (Figure S6).

Discussion

Our systematic review estimated the prevalence of HT among adults in SSA at 27.09% (95% CI: 24.22%–29.96%), with substantial inter-study heterogeneity and marked regional disparities. The highest prevalence was observed in West Africa (30.66%) and Central Africa (32.75%), whereas East Africa (24.54%) and Southern Africa (22.43%) reported significantly lower levels. The extremely high heterogeneity (I^2 > 99%) observed across studies limits the interpretability of the pooled prevalence estimate. Subgroup analyses and prediction intervals were used to explore variability and provide a more nuanced understanding of the burden of hypertension across different settings. This geographic variability likely reflects differences in urbanisation [78, 79], access to healthcare [7], dietary patterns, sedentary lifestyles [80], as well as genetic and environmental determinants [81].

Comparison with the Global Burden of Disease (GBD) 2021 estimates shows general concordance, with the GBD reporting a HT prevalence of approximately 30–32% among adults aged 30–79 years and highlighting little progress in SSA over the past three decades, in contrast to other regions [82]. However, whereas the GBD relies on older data and global modelling, our meta-analysis provides recent, region-specific estimates, revealing marked regional disparities often obscured in global analyses. This contribution is particularly relevant given that SSA continues to exhibit the lowest rates of HT screening, treatment, and control, with control rates below 10% in several countries [82]. By providing granular and up-to-date data, our study enhances understanding of regional dynamics and can inform more targeted interventions adapted to local health system realities. Previous systematic reviews have provided important regional benchmarks. Chen et al. [83] reported a pooled hypertension prevalence of 30.5% (95% CI: 28.4–32.6%) among adults in 26 SSA countries, which is broadly similar to our estimate but based on a slightly different study selection and analytical scope. Additionally, a recent meta-analysis focused on West African populations reported a pooled prevalence of 33.8% (95% CI: 29.3–38.7%) [84]. Earlier work, including analyses of uncontrolled hypertension among people with comorbidities, highlights the persistent burden of hypertension and related management gaps in SSA, though these studies differ in focus [22]. In contrast, our analysis focused exclusively on community-based studies conducted in SSA between 2010 and 2025. Variations in population age

Table 1 Characteristics of studies included on hypertension prevalence among adults in Sub-Saharan Africa

Author (Year)	Sub-region	Country	City/Village	Study setting	Study periode	Study populationage	Total	Number of participants with hypertension	Prevalence (%)	JBI score (%)
Abebe et al. (2015)	Eastern	Ethiopia	Gondar town and Dabat rural kebeles	Mixed	2010–2014	≥ 35 years	2141	598	27.93	88
Adeke et al. (2024)	Western	Nigeria	Abuja Municipal, Gombe Municipal, Gusau, Onitsha, Uyo, Ibadan-North, Gwagwalada, Akko, Bungudu, Oyi, Nsit Ubium, Akinyele LGAs	Mixed	2015–2019	≥ 18 years	3782	1080	28.56	88
Anjulo et al. (2021)	Eastern	Ethiopia	Areka	Urban	2015–2019	≥ 25 years	576	110	19.10	75
Anto et al. (2023)	Western	Ghana	Ablekuma North	Rural	2020–2025	18–80 years	1288	237	18.40	75
Asemu et al. (2021)	Eastern	Ethiopia	Addis Abeba	Urban	2015–2019	≥ 18 years	3560	996	29.24	88
Chelo et al. (2023)	Central	DRC	Goma	Urban	2020–2025	≥ 18 years	4046	809	20.00	88
Chuka et al. (2020)	Eastern	Ethiopia	Arba Minch	Mixed	2015–2019	25–64 years	3346	633	18.92	100
De-Ramirez et al. (2010)	Eastern	Malawi, Rwanda, Tanzania	Mwandama, Mayonga, Mbola	Rural	2010–2014	≥ 18 years	1485	325	21.89	88
Dereje et al. (2020)	Eastern	Ethiopia	Hosanna town	Urban	2015–2019	≥ 18 years	627	108	17.22	75
Desormais et al. (2019)	Western	Benin	Taneve	Rural	2015–2019	≥ 25 years	1777	584	32.86	88
Dosoo et al. (2019)	Western	Ghana	Kintampo North	Mixed	2015–2019	≥ 18 ans	2555	717	28.06	75
Geleta et al. (2019)	Eastern	Ethiopia	Nekemte town	Urban	2015–2019	18–80 years	705	246	34.89	88
Guwatudde et al. (2015)	Eastern	Uganda	Central, Northern	Mixed	2010–2014	18–69 years	3906	1033	26.45	75
Helelo et al. (2014)	Eastern	Ethiopia	Durame Town	Urban	2010–2014	≥ 31 years	518	116	22.39	88
Houinato et al. (2012)	Western	Benin	Benin	Mixte	2010–2014	25–64 years	6786	1894	27.91	88
Isara et al. (2015)	Western	Nigeria	Edo state, rural communities	Rural	2010–2014	≥ 18 years	845	319	37.75	88
Jobe et al. (2024)	Western	Gambia	National	Mixed	2015–2019	≥ 35 years	9188	4311	47.00	88
Jorgensen et al. (2020)	Eastern	Tanzania	Zanzibar	Urban	2010–2014	25–64 years	2146	813	33.50	88
Joshi et al. (2014)	Eastern	Kenya	Nairobi	Urban	2010–2014	18–64 years	2061	258	22.80	88

Table 1 (continued)

Author (Year)	Sub-region	Country	City/Village	Study setting	Study periode	Study populationage	Total	Number of participants with hypertension	Prevalence (%)	JBI score (%)
Kotwani et al. (2013)	Eastern	Uganda	Kakyerere parish (Mbarara)	Rural	2010–2014	≥ 18 years	2252	354	15.72	88
Kumma et al. (2021)	Eastern	Ethiopia	Wolaita, southern Ethiopia	Mixed	2015–2019	25–64 years	2483	818	32.94	88
Macia et al. (2015)	Western	Senegal	Dakar	Urban	2010–2014	≥ 20 years	600	165	27.50	88
Matsuzaki et al. (2020)	Southern	Mozambique	Manica and Sofala	Mixed	2015–2019	≥ 20 years	4101	513	15.80	88
Mbouemboue et al. (2016)	Central	Cameroon	Ngaoundere	Urban	2010–2014	≥ 18 years	700	143	20.43	88
Mekonene et al. (2023)	Eastern	Ethiopia	Addis Abeba	Urban	2020–2025	18–64 years	600	133	22.17	63
Mohamed et al. (2018)	Eastern	Kenya	–	Mixed	2015–2019	18–69 years	4433	1270	28.65	75
Mondo et al. (2013)	Eastern	Uganda	Kasese	Urban	2010–2014	25–64 years	611	172	21.44	88
Mphekgwana et al. (2020)	Southern	South Africa	Polokwane, Gamolesa, Seshego, Limpopo	Mixed	2015–2019	≥ 18 years	2293	454	21.33	75
Musung et al. (2021)	Central	DRC	Lubumbashi	Mixed	2015–2019	≥ 18 years	6708	2251	33.56	88
Njonnou et al. (2024)	Central	Cameroon	Dshang	Mixed	2020–2025	≥ 21 years	706	407	57.65	88
Odili et al. (2020)	Western	Nigeria	Abuja, Owagwalada, six zones	Mixed	2015–2019	≥ 18 years	4192	1597	38.10	75
Olack et al. (2015)	Eastern	Kenya	Soweto, Gatwikira, Mashimoni (Kibera)	Urban	2010–2014	35–64 years	1528	418	27.36	88
Oladapo et al. (2010)	Western	Nigeria	Egbeda district	Rural	2010–2014	18–64 years	2000	415	20.80	75
Olamoyegun et al. (2016)	Western	Nigeria	10 semi-urban communities	Mixed	2010–2014	≥ 18 years	750	416	55.47	88
Omar et al. (2020)	Eastern	Sudan	Gadarif	Mixed	2015–2019	≥ 18 years	600	245	40.83	88
Onwuchekwa et al. (2012)	Western	Nigeria	Kegbara	Rural	2010–2014	≥ 18 years	1078	197	18.27	88
Opoku et al. (2020)	Western	Ghana	10 regions: Western, Central, Greater Accra, Volta, Eastern	Mixed	2010–2014	≥ 18 years	12,151	5093	11.48	88
Opreh et al. (2021)	Western	Nigeria	South West Nigeria	Rural	2020–2025	≥ 20 years	1012	461	45.55	88

Table 1 (continued)

Author (Year)	Sub-region	Country	City/Village	Study setting	Study periode	Study populationage	Total	Number of participants with hypertension	Prevalence (%)	JBI score (%)
Osunkwo et al. (2020)	Western	Nigeria	Makundi town, Katsina village	Mixed	2015–2019	≥ 18 years	1265	450	35.57	88
Pengpid et al. (2020)	Southern	Zambia	National	Mixed	2015–2019	18–69 years	4302	852	18.90	88
Pengpid et al. (2022a)	Central	RCA	Bangui and Ombella M'poko	Mixed	2015–2019	25–64 years	3265	1892	42.05	88
Pengpid et al. (2022b)	Eastern	Sudan	–	Mixed	2015–2019	18–69 years	7226	2747	38.02	75
Pires et al. (2013)	Central	Angola	Bengo province and Dande municipality	Mixed	2010–2014	18–64 years	1464	322	23.02	88
Price et al. (2018)	Southern	Malawi	Korongo district, Lilongwe City	Mixed	2010–2014	≥ 18 years	28,891	4096	14.18	63
Ratovoson et al. (2015)	Southern	Madagascar	Moramanga	Mixed	2010–2014	≥ 18 years	7631	2170	28.44	88
Solomon et al. (2023)	Eastern	Ethiopia	Gurage zone, SW	Mixed	2020–2025	≥ 18 years	635	140	22.05	88
Soubeiga et al. (2017)	Western	Burkina Faso	Ouagadougou	Urban	2010–2014	≥ 18 years	4629	411	18.00	75
Tateyama et al. (2022)	Southern	Zambia	Mumbwa district, Central province	Mixed	2015–2019	25–64 years	689	252	36.57	88
Tesfaye et al. (2019)	Eastern	Ethiopia	Amhara	Mixed	2015–2019	≥ 18 years	1405	160	11.39	63
Teshome et al. (2022)	Eastern	Ethiopia	Dabat and Gondar zuria	Rural	2020–2025	≥ 18 years	1177	218	18.52	75
van de Vijver et al. (2013)	Eastern	Kenya	Nairobi, Vivaudavi, Karogocho	Urban	2010–2014	≥ 18 years	5190	659	12.29	75
Zekewos et al. (2019)	Eastern	Ethiopia	Bona	Rural	2015–2019	≥ 25 years	1952	200	21.82	100

structures, urban–rural composition, and hypertension definitions may partly explain differences in pooled estimates. Stratifying by data collection period showed no statistically significant differences (Q = 3.15; p = 0.207), but mean prevalences rose from 24.3% (2010–2014) to 29.2% (2015–2019) and 29.2% (after 2020), suggesting a gradual increase. This trend, though not statistically significant, may reflect entrenched cardiovascular risk factors amid underdeveloped prevention policies, dietary changes, and declining physical activity [85]. Together, these comparisons suggest that HT prevalence in SSA has remained high over the past decade, supporting a trend of sustained burden rather than marked reduction over time. Overall, these comparisons indicate that while regional estimates remain broadly consistent with prior reviews, our analysis provides updated and regionally disaggregated data, supporting more targeted public health planning and policy interventions.

The Egger regression test, applied after logit transformation of proportions, did not reveal significant publication bias (p = 0.5687), supporting the internal validity and robustness of the aggregated results. Overall, this study

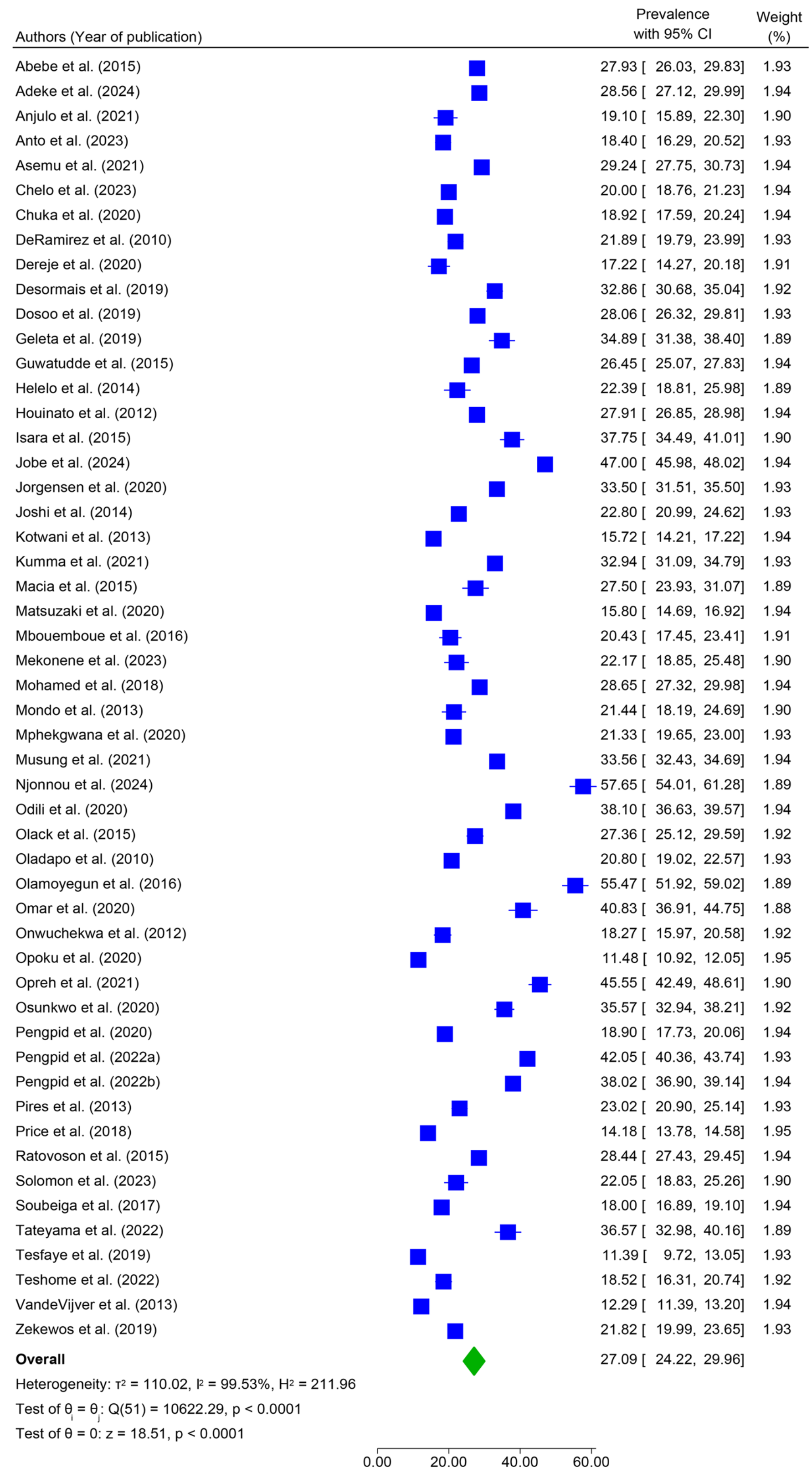

Fig. 3 Forest plot of the meta-analysed prevalence of hypertension among adults in sub-Saharan Africa

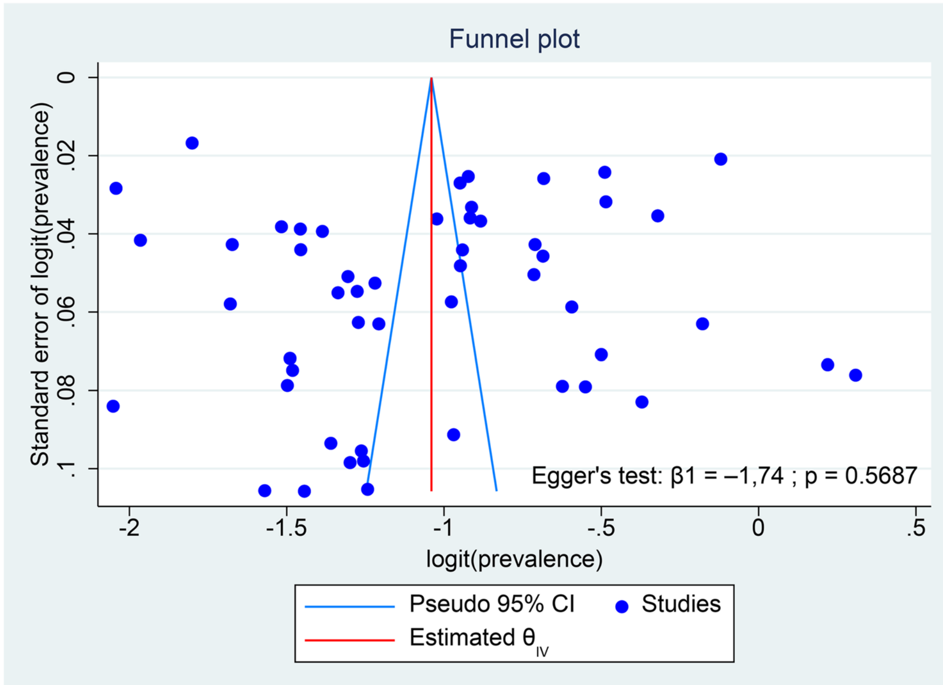

Fig. 4 Funnel plot assessing publication bias in the prevalence of hypertension among adults in sub-Saharan Africa

highlights a substantial and growing epidemiological burden of HT in SSA, characterised by uneven geographic distribution, complex socio-spatial dynamics, and insufficient public health responses [22, 51, 86]. It underscores the need for integrated, multisectoral interventions addressing social, environmental, and behavioural determinants [13]. Strengthening primary healthcare, community education, equitable access to screening services, and promotion of healthy lifestyles should form the cornerstone of any regional strategy aimed at mitigating the HT burden in the coming years [87].

Our findings on associated factors confirm determinants already reported in global and SSA literature. Advanced age, male sex, urban residence, family history, diabetes, overweight/obesity, as well as behaviours such as alcohol consumption, smoking, physical inactivity, and low fruit and vegetable intake, were strongly associated with HT. The strong association between advanced age and HT observed in our meta-analysis is consistent with the findings of Chen et al. [83], who reported a marked increase in HT prevalence after the age of 50 years. This age gradient likely reflects cumulative exposure to behavioural factors, progressive vascular stiffening, and age-related metabolic changes, underscoring the importance of early-life and mid-life prevention strategies to curb HT burden later in adulthood. These factors reflect the complex interplay between genetics, lifestyle, and environment, highlighting key targets for prevention and disease management. Among modifiable determinants, overweight and obesity remain major contributors, often exacerbated by sedentary behaviour, diets high in sodium and saturated fats and low in fibre, fruits, and vegetables, as well as excessive alcohol consumption [4, 8]. Diabetes, frequently associated with metabolic syndrome, further increases HT risk due to its vascular and metabolic effects, while smoking and chronic stress contribute to endothelial dysfunction and excessive sympathetic activation. Non-modifiable factors, such as advanced age and family history, emphasise the role of genetic susceptibility, particularly in certain SSA populations, which can interact with environmental and behavioural determinants to accelerate the onset and progression of HT.

Collectively, these findings highlight the importance of integrated preventive approaches combining lifestyle interventions, early screening, and management of metabolic comorbidities to effectively reduce the HT burden in SSA. Promotion of balanced diets, reduction of alcohol intake, smoking cessation, and encouragement of regular physical activity constitute essential levers for both primary and secondary HT prevention [4, 82].

Several limitations should be acknowledged. The very high heterogeneity across studies (I^2 > 99%) indicates substantial variability in study populations, blood pressure measurement procedures, and HT definitions. A key source of this heterogeneity is variation in blood pressure measurement protocols. Critical details—including the number of readings, type of device (manual versus automated), rest period before measurement, and observer training—were inconsistently reported. Such methodological variability is a well-recognised source of bias in HT prevalence research and likely contributed to the observed inter-study heterogeneity. Small sample sizes and potential selection biases in some studies may have further affected the precision. Additionally, the predominance of cross-sectional designs limits causal inference regarding factors associated with HT.

Although studies were drawn from multiple subregions of sub-Saharan Africa (SSA), geographic coverage was incomplete and uneven. Of approximately 46 countries in the region, only 21 were represented, with a substantial proportion of evidence coming from a few countries, particularly Ethiopia and Nigeria. Data were concentrated mainly in West and East Africa, while Central Africa and parts of Southern Africa were sparsely represented or had no data. This imbalance reflects disparities in research capacity and data availability, limiting the external validity of the pooled estimate.

All included studies were community- or subnational population-based surveys conducted in urban, rural, or mixed settings. No nationally representative household surveys, such as DHS or WHO STEPS, were identified. Consequently, the pooled prevalence reflects the distribution of available evidence rather than a population-based regional estimate. Reliance on geographically clustered and predominantly local studies may introduce selection bias and could result in either over- or underestimation of the true regional burden, particularly in settings with marked urban–rural, socioeconomic, and healthcare access disparities.

Furthermore, pooled estimates for HT awareness, treatment, and control could not be generated due to inconsistent reporting and heterogeneous definitions, limiting assessment of the HT care cascade. Despite an exhaustive search, quantitative synthesis was constrained by substantial methodological heterogeneity and incomplete reporting of associated factors, which limited the scope and depth of subgroup analyses. Accordingly, the pooled prevalence should be interpreted as a summary indicator of HT magnitude in settings where data are available, rather than as a precise population-wide estimate for SSA.

These findings have important public health implications for SSA. The high prevalence of HT and the documented regional disparities highlight the urgent need for targeted screening programmes adapted to local contexts. Preventive strategies focusing on a healthy diet, regular physical activity, reduction of alcohol consumption, and smoking cessation are essential to mitigate cardiovascular risk in vulnerable populations [79, 88]. Incorporating metabolic and familial risk factors could improve the identification of the most susceptible groups, as several studies have demonstrated their key role in the onset and severity of HT in SSA [7].

From a policy perspective, our findings reinforce the need to strengthen primary healthcare systems, expand access to HT screening and treatment, and promote high-impact interventions such as task-shifting or integration of HT care into community health services [22, 88, 89]. Implementing multisectoral approaches that integrate nutrition, urban planning, health education, and non-communicable disease prevention is essential to sustainably reduce the HT burden in SSA [88, 90–92].

For future research, longitudinal and nationally representative studies are needed to better characterise causal determinants of HT and temporal trends across SSA. Standardisation of blood pressure measurement protocols is also essential to improve comparability between studies. In addition, evaluating the effectiveness of community-based and individual-level interventions adapted to local contexts, and examining regional and urban–rural differences using harmonised data and advanced analytical approaches, would help inform more targeted strategies for HT prevention and control.

Conclusion

These findings underscore the urgent need for integrated, multisectoral interventions combining strengthened primary healthcare, targeted screening, promotion of healthy lifestyles, and health education, tailored to local contexts and at-risk populations. Moreover, longitudinal and nationally representative studies are required to better elucidate causal determinants of HT and to evaluate the effectiveness of interventions in diverse SSA settings. Overall, this study provides essential evidence to guide prevention and management policies for HT in SSA and to reduce its impact on cardiovascular morbidity and mortality across the region.

Abbreviations

95% CI	95% confidence interval
HT	Hypertension
OR	Odds ratio
SSA	Sub-Saharan Africa

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

PKK conceived the study, designed the protocol, and coordinated data extraction. EKM contributed to the study design, data collection, and verification of extracted data. OM supervised the methodological development, conducted the statistical analyses, and drafted the initial manuscript. JMM participated in data acquisition, literature search, and quality assessment. JPMB contributed to data interpretation and critical revision of the manuscript. GKL assisted in data extraction and contributed to the writing of the background and discussion sections. KKN contributed to the literature review and supported data verification. GMKB assisted with methodological validation and manuscript editing. JBSZZK participated in the interpretation of findings and manuscript revision. MB contributed to data curation and provided technical input for the analysis. DNN participated in study coordination, data quality control, and manuscript review. BJHvB provided senior supervision, contributed to the interpretation of results, and critically revised the final manuscript for important intellectual content. All authors read and approved the final version of the manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author upon reasonable request.

Declarations

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Not applicable.

Consent for publication
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